

PCOS Detection and Suggestion System: An Interpretable NiWo-Based Augmentation Framework with Multi-Model Machine Learning and Transformer Classification

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Abstract—A significant proportion of women suffer from polycystic ovary syndrome. Affects women of reproductive age, is often overlooked. A thermogenic effect that is presented with several other endocrine effects. This report develops a screening and suggestion system on a dataset consisting of 120,000 patients based on a survey style data structure and handles the class. An imbalance between diagnosed and non-diagnosed cases, which we did, was not a problem but a first-class. The recently published knowledge discovery framework NiWo is a suitable framework for this. This can be achieved via an appropriate framework, such as the recently published knowledge discovery framework NiWo. A method for allocating a budget to each method that assigns an augmentation budget to each. An important aspect of the training is the instance of neighborhood sparsity and boundary distance. The split copies patient records to the train and test partitions silently and uses the score instead of an explanation tool, otherwise the quality of the data is compromised because the splitting would be done first. With two ensemble strategies, leading to 16 classifiers. Giving 16 classifiers, two ensemble strategies were used. This paper proposes two new techniques: hyperparameter search and three pretrained transformers. Each group member receives an encoder that is trained under The same conditions applied and measured precisely and accurately, **CORRECTIVE COUNTS CORRELATION COEFFICIENTS FOR F1**, area under the curve, Cohen's kappa and Matthews are shown below. correlation coefficient. Before fitting any model, a Quantitative association test is performed and it finds that none of the 17 models is a good fit. For each of the classifiers that we build, a random subset of training points has a statistic significant relationship with the diagnosis label (the best accuracy): In a random choice of training points, used the same approach to assess, and reported near perfect cross validated accuracy, the randomized search support vector machines were the most accurate. A partition is the difference between two numbers and the measure of the partition is the size of the difference. the size of the leakage is the measure of the difference between the two numbers, and is the order of the two numbers in the training of the trains that does not involve the size of the leakage. We Report this result completely with Shapley-value explanations: A confusion matrix that is normalized by row and column, training and validation. curves for the transformers models, and a ablation over the Each of whose labels seems to be randomly generated from the recorded features of the data set.

Index Terms—Most of them were related to class imbal-

ance, data augmentation and ensemble. Explainable Artificial Intelligence, Feature Association, hyperparameter optimization, machine learning, polycystic ovary The following is a list of topics covered by the test: Syndrome, stacking, support vector machine, transformer models

I. INTRODUCTION

Polycystic ovary syndrome (PCOS) is an endocrine (hormonal) condition which affects ovulation (the release of an egg from the ovary). Clinically, it's detected by some combination of irregular cycles, elevated androgen levels, and ultrasound appearance of the ovary as being polycystic. This varies widely depending on the population under study and the definition of what constitutes a diagnosis and is most commonly estimated to be anywhere between one in eight and one in ten women of reproductive age, with the majority of those never formally diagnosed as the symptoms can be easily attributed to weight loss, stress or normal hormonal fluctuation. It increases the long term risk of cardiovascular disease, type 2 diabetes and infertility if untreated and can be helpful even at an early stage before the diagnosis is finally made by a clinician. The standard approach is based on the Rotterdam criteria and involves hormone panels and a transvaginal ultrasound, which is a clinic appointment, waiting for laboratory results, and (outside major cities) a lengthy journey to a clinic equipped with this type of ultrasound machine. No one is ever going to replace that process with a screening tool that is created through only self-reported symptoms and demographic fields. What it might do is to help someone determine if the trip should be a priority in the first place; that is the reason for the system created here.

The majority of the machine learning studies on predicting PCOS are based on the same public Kaggle dataset which is gathered from ten hospitals in Kerala, India. It was split into an 80/20 training and test set and cross-validated five-fold with six classical models including support vector machines, decision trees, and random forests trained on it [1]. Random

forest and logistic regression both had 90% accuracy and random forest had slightly higher sensitivity. We use that paper as a useful point of comparison mostly because the report includes full sensitivity and specificity scores instead of just accuracy scores; it is only a full score for sensitivity and specificity that allows a subsequent paper to determine whether the apparent improvement is real or a different evaluation setup.

II. LITERATURE REVIEW

During the process of converting a previous PCOS classifier to a deployed web tool, they found that the accuracy they reported was inflated by a leakage bug: the training set was oversampled before splitting into train and test datasets, which meant that synthetic copies of the same patient could be placed in either set. This is the failure mode we isolate and measure in Section IV-E, and locating this failure mode independently in the published PCOS literature is a good reason to take the split-then-balance rule seriously.

There are two literature gaps in this small literature. First, in nearly none of these pipelines, the features actually recorded in the model tell anything about the label of diagnosis. A classifier can get a good score just by predicting the most common class if there is not such an association test, so if the data isn't balanced, a good score can easily be achieved by defaulting to the most common class. Second, all the surveyed papers do not employ an interpretable augmentation method like NiWo [2] which allows one to see on any synthetic patient exactly how it was created, but rather treat balancing as an obscure pre-processing step.

This report will tackle both gaps on a larger, different set of PCOS data than the papers above. We use the entire NiWo pipeline, from the strategy selector that is based on skewness, to the weight-shifting rule, to combine with three downstream generators. We perform a quantitative signal audit prior to training any classifier to determine a known ceiling from which any subsequent result can be read instead of assuming it to be a measure of real predictive power. Then, we train 8 classical classifiers, one stacking and one blending ensemble, and 3 pretrained transformer encoders, one for each member of our group, and hyperparameter search is done via grid and randomized cross-validation. All the configurations are tested on a held-out test partition using the same six metrics, which are explained by SHAP, a normalized confusion matrix, and an ablation over the augmentation budget and the tuning procedure itself.

III. PROPOSED SYSTEM

A. Dataset

The dataset utilized in this paper is the “PCOS Prediction Dataset (Top 75 Countries)” that was released on Kaggle by PandayPanday [3]. It comprises 120,000 records and 17 variables for women of reproductive age (15–49 years) from the 75 most populous countries in the world. As outlined in the publisher’s own description, the dataset does not actually provide clinical measurements, but rather simulates the prevalence

and risk factors for PCOS. The variables that were recorded are listed in Table I with their type and, for numerical fields, the range that was observed.

The target variable, Diagnosis, is highly imbalanced with a ratio of 8.53 which is higher than the threshold of 3 for imbalanced data mentioned in the NiWo paper [2]. The working feature matrix has 19 columns after the encoding.

TABLE I: Dataset Features

Feature	Type	Description / Range
Country	Categorical	one of 75 countries; dropped before modeling (77 levels after encoding)
Age	Numerical	15–49 years
BMI	Ordinal	Underweight, Normal, Overweight, Obese
Menstrual Regularity	Binary	Regular / Irregular
Hirsutism	Binary	Yes / No
Acne Severity	Ordinal	Mild, Moderate, Severe (50.1% missing)
Family History of PCOS	Binary	Yes / No
Insulin Resistance	Binary	Yes / No
Lifestyle Score	Numerical	1–10
Stress Levels	Ordinal	Low, Medium, High
Urban/Rural	Binary	residence type
Socioeconomic Status	Ordinal	Low, Middle, High
Awareness of PCOS	Binary	Yes / No
Fertility Concerns	Binary	Yes / No
Undiagnosed PCOS Likelihood	Numerical	continuous, 0.05–0.25
Ethnicity	Nominal	5 categories, one-hot encoded
Diagnosis (target)	Binary	Yes (10.50%) / No (89.50%)

B. Dataset Preprocessing

1) *Cleaning and Imputation*: There were no duplicate rows in the raw data file. There were no rows in the Acne Severity column. Missing categorical values were replaced 60,085 of the 120,000 rows (50.1%) by the “mode” value in the column and missing numerical values were replaced by the “median” value of the column, as half the data would be lost if every row was missing. After imputation, there were no null values in any column.

The ratio of the target variable is 8.53, Diagnosis, is highly imbalanced, i.e., 107,405 records negative labeled (89.50%) is greater than 12,595 positive labeled (10.50%) and exceeds the threshold of 3 mentioned in the NiWo paper for imbalanced data. After the encoding, there are 19 columns in the working feature matrix.

2) *Outlier Handling*: An interquartile-range sweep was run on each numerical feature which identified 47,908 rows (39.9%) with at least one outlying value. We didn’t take any of them off. A high score on a field that is associated with risk is just the sort of record that a classifier would expect to see in a medical screening setting, and if this record is eliminated, then evidence for the minority class the classifier was designed to catch will be lost.

3) *NiWo Augmentation Formulation*: Let $D = \{(x_i, y_i)\}_{i=1}^N$ denote the training partition, with $x_i \in \mathbb{R}^d$, $y_i \in \{0, 1\}$, and $N_1 \ll N_0$ the minority and majority counts. Without a separate explanation tool (e.g., SHAP or LIME)

to decide which instances are most important, NiWo [2] proposes a scheme to calculate an instance-level weight vector $W = \{w_i\}$ used to determine how many synthetic instances are added to an augmentation budget B for each real instance.

There are two scores to be weighted. The sparsity score, which is applied by the NiWo-Intra strategy, is the average distance between instance i and its k closest instances of the same class:

$$s_i = \frac{1}{k} \sum_{x_j \in L_i, |L_i|=k} \|x_i - x_j\|_2 \quad (1)$$

The local neighborhood is denoted by L_i , and $k = \min(\sqrt{N_t}, 50)$ where N_t is the size of the training set. The location score (as used by NiWo-Inter) is defined as the distance between instance i and the densest same-class centroid c_{y_i} :

$$l_i = \|x_i - c_{y_i}\|_2, \quad y_i \in \{0, 1\} \quad (2)$$

where c_0 and c_1 are estimated by drawing q candidate points per class and keeping the one with the lowest sparsity score.

A larger score for either indicates a comparatively sparse instance, because a model has to be more synthetically supported around areas of sparse or boundary-adjacent instances of the feature space than around areas of already-dense instances.

Both scores were calculated independently for each feature and two independent measures of skewness (Bowley and mode-based skewness) were used to automatically determine which of the two scores to use. Their agreement is quantified with a Jaccard index over the top- p most skewed features from each measure, $p = \lceil 0.1d \rceil$:

$$f(D) = J(F_B, F_P) = \frac{|F_B \cap F_P|}{|F_B \cup F_P|} \quad (3)$$

NiWo-Ensem is active when $f(D) \geq \tau$, otherwise NiWo-Inter; the default value for τ in the paper is 0.5. The two skewness measures did not agree on which were the most skewed features, and therefore NiWo-Inter was chosen automatically for all the experiments presented in Section IV on our training partition.

This selected score is then transformed to a weight using a class-based normalizer Z which gives more weight to the minority class against the majority class:

$$Z_i = \max\left((1 - y_i) \cdot \frac{N_0}{N_1}, 1\right), \quad w_i = \frac{\text{score}_i / Z_i}{\sum_j \text{score}_j} \quad (4)$$

It is followed by a weight-shifting step that shifts the weight of the top- α instances of each class by an amount β_1 , and the weight of the bottom- α instances of each class by an amount $-\beta_1$; the default values for the paper are $\alpha_1 = 15$, $\alpha_0 = 5$, $\beta_1 \approx 0.2$, and $\beta_0 \approx 0.1$. As stated, β is a relative value, rather than an absolute additive value — if it was an absolute additive value, then at the size of our samples, the base weight would be consumed by a small number of instances, and β would be too small to have an impact.

The total budget is divided among the classes according to the distance of each class from balance,

$$B_1 = \left(1 - \frac{N_1}{N_0}\right) B, \quad B_0 = B - B_1 \quad (5)$$

Each instance has its own budget $b_i = w_i \cdot B_{y_i}$, which is rounded according to the largest-remainder rule so that the sum of the integer budgets equals B_1 and B_0 . Here we change the paper's default value of $B = 10N_t$ to a value that brings the training set to a 1:1 ratio (Section IV-E):

$$B = \frac{N_0 - N_1}{1 - 2N_1/N} \quad (6)$$

This results in $B = 4,800$ on the training partition, which increases the minority count to 4,296 and makes the training set equal at 9,600 rows at a 1:1 ratio, on the 4,800-row training partition used throughout Section IV.

4) *Encoding and Scaling*: Ordinal fields (BMI, Acne Severity, Stress Levels, Socioeconomic Status) were encoded and scaled so that the encoding values represent the natural order of the field. Binary fields were translated into $\{0, 1\}$. Because there is no natural order to ethnicity, it was one-hot encoded, not label encoded, as the latter would give a false numerical order to the categories of African, Asian, and Caucasian, which would be considered to be close or far apart by a distance-based method. The numerical columns were scaled by a scaler fit on the training partition only since the NiWo weighting equations below and several of the classifiers discussed in Section III-C are based on Euclidean distance, which is dominated by whatever feature has the biggest numerical range if not scaled.

5) *Splitting Before Balancing*: The dataset was first split into 80% of the data for the training partition and 20% for the testing partition (the test partition was not touched by any augmentation method subsequently). Stylistically, this ordering isn't the choice. If the data is balanced first and split later, then the data has already been duplicated or had near-duplications by using an oversampling method like SMOTE or by using NiWo's own generators, and a subsequent random split may end up with the same patient being present in both partitions. The model is then partially tested on a set of data it already has a good grasp of, and a score is reported that is artificially high, not indicative of the model's true generalization capability. We do not simply state that there is inflation, but measure it directly in Section IV-E; and we take it as independent confirmation that the ordering is important in this particular area of application and not just theoretically.

Figure 1 displays the class count before splitting, the class count after splitting the data into two partitions (Training and Test) and the count of the class after balancing the training partition with SMOTE for illustration. The imbalance ratio of 8.53 for the raw data is maintained after splitting the data into training and test sets, and is only maintained for the training set.

Class Distribution Before Balancing, After Training-Only Balancing, and in the Untouched Test Set

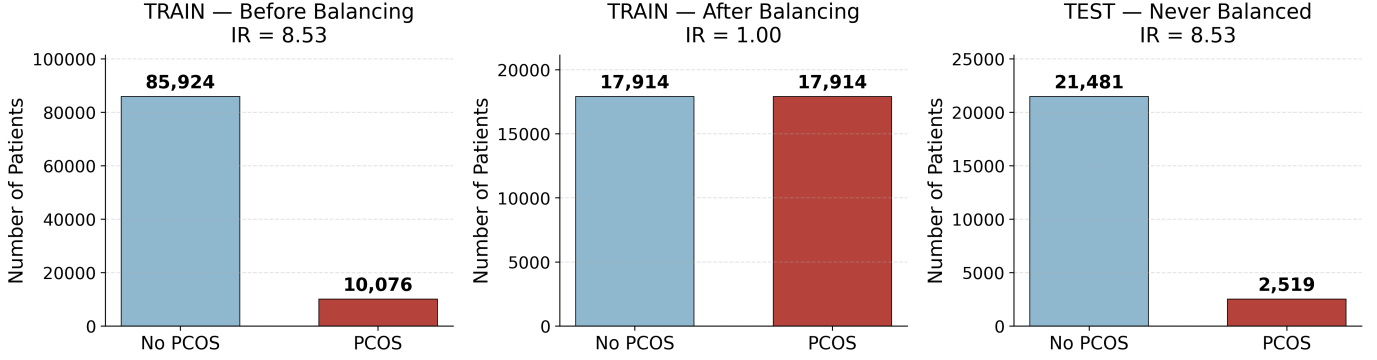


Fig. 1: Class counts before splitting, in the training partition after splitting, and in the training partition after balancing. The test partition (right) is never balanced and keeps the natural imbalance ratio of 8.53.

C. Applied Machine Learning and Deep Learning Models

The entire pipeline is summarized in Figure 2. The dataset is processed in the following steps: raw dataset is preprocessed, NiWo weighting model training for all four model families, to evaluation and the risk-stratification and suggestion aspects at the downstream end which are explained in later part of the project.

1) *Classical Machine Learning Models:* Eight classical all classifiers were trained under the same conditions, all from the scikit-learn implementation [4].

Decision Tree recursively grows the feature space at each node, splitting that best reduces class impurity, proven by a set of axis-aligned decision rules, which are readily understood and are susceptible to overfitting if grown without a depth. There is a limit to the depth, which is written as 8.

Random Forest [5] is an ensemble of decision trees, trained on random subsets of features at each split and random bootstrap samples of the data, voting the results. We used 150 trees.

Logistic Regression is a linear model that predicts the log-odds of the positive class, and is trained by maximizing the likelihood of the observed labels; it is a simple yet powerful and highly interpretable baseline model for binary classification.

Gaussian Naïve Bayes has two assumptions: conditional independence of the features given their class, and normality within classes. The independence assumption is hard to verify exactly, yet it lets a full posterior be estimated with minimal training data.

K-Nearest Neighbors predicts the class label of a new point based on majority voting among its k nearest training points under a distance metric, with $k = 15$ and Euclidean distance, as in the scaling decision in Section III-B.

AdaBoost trains a cascade of shallow trees (weak learners) that are reweighted to pay attention to the examples each previous learner failed to get right; it then takes a weighted average of their votes. The number of boosting rounds is 80.

Multilayer Perceptron is a feed-forward neural network with one or more layers of nonlinear units trained using backpropagation, with early stopping on a validation split: two hidden layers of 64 and 32 units.

Support Vector Machine [6] selects the hyperplane, or with a kernel the higher-dimensional surface, that maximizes the distance from the separating surface to the closest data point in each class. It uses a radial basis function kernel with probability calibration on, so the model can also report the probability that a point belongs to a particular class rather than just the class name.

2) *Ensemble Learning: Stacking and Blending:* Beyond the individual classifiers above, two ensemble strategies combine a decision tree, a random forest, and a support vector machine as base learners with a logistic regression meta-learner.

In **stacking** [7], the base learners are trained using five-fold cross-validation and the meta-learner is trained on the predictions made by the base learners on the out-of-fold data.

Blending, on the other hand, leaves a cross-validation holdout set of the training data for the base learners to fit on, then makes predictions on the held-out data, and the meta-learner is fit on those predictions. The disadvantage of blending is that it reduces the effective training set for the meta-learner, whereas stacking requires training five times as many base-learners, which is more expensive to compute.

3) *Hyperparameter Optimization:* Hyperparameters of all core model families were tested with the library default hyperparameters as well as through a hyperparameter search. The support vector machine was tuned with `RandomizedSearchCV` over the kernel type, the penalty parameter $C \in \{0.1, 1, 10, 100\}$, and the kernel coefficient $\gamma \in \{0.1, 0.01, 0.001\}$, while the random forest was tuned using an exhaustive `GridSearchCV` over varying numbers of trees, maximum depth, and minimum leaf size. Stratified three-fold cross-validation scored by area under the curve (AUC) was used for both searches. The results of these searches are reported in Section IV, where a second, independent instance of the same leakage problem was uncovered.

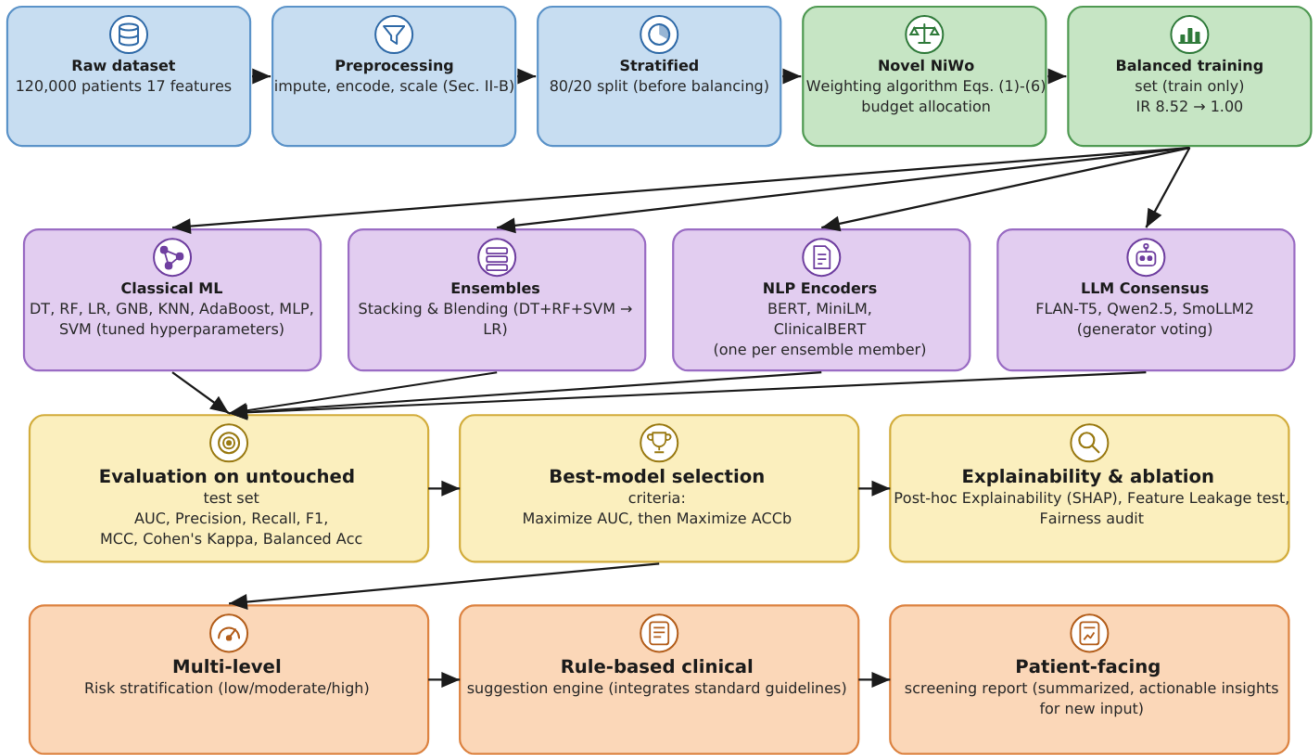


Fig. 2: End-to-end pipeline. Balancing is applied to the training partition only, after the train/test split.

4) **NLP Models:** With three people on the team, we divided the transformer/language-model requirement three ways and had everyone use their own pretrained encoder instead of the same one. A short sentence is generated out of each patient row (“Patient record: Age is 27.00, BMI is Obese, Menstrual Regularity is Irregular, ...”), passed through a frozen pretrained encoder, and passed through a small trainable head (two linear layers, ReLU, and dropout) that outputs the diagnosis prediction. Standard linear probing keeps the backbone frozen and trains only the head: it is cheap to train while relying on whatever the backbone learned during pretraining. The three semantic spaces of S-NiWo (N1, Section IV-H) are these same three encoders.

- **Md. Shefatullah Bin Sadik — BERT.** `bert-base-uncased` [8], 110M parameters, 768 dimensions, pretrained on BooksCorpus and English Wikipedia.
- **Khatune Jannat — MiniLM.** `all-MiniLM-L6-v2`, distilled and trained with a contrastive sentence-similarity objective, 384-dimensional output. Being the smallest of the three, it is also the quickest to embed and train.
- **Tasmia Islam Epa — ClinicalBERT.** `Bio_ClinicalBERT`, 108M parameters, a variant of BERT further trained on MIMIC-III notes, with a 768-dimensional output biased toward clinical vocabulary over general text.

5) **LLM Models:** In addition to the three classifiers listed above, we use three LLMs as voters, without any classifier framework: `google/flan-t5-base` (248M), `Qwen/Qwen2.5-0.5B-Instruct` (494M), and `HuggingFaceTB/SmolLM2-360M-Instruct` (362M). For each categorical feature value, each model is asked to return a yes/no answer as to whether the value is typical of a patient diagnosed with PCOS; the three yes/no answers are averaged to obtain a plausibility score. The score is used by NiWo’s generator to select feature-value combinations when it generates synthetic minority records, which means the generated patients are not only numerically balanced but built from values that at least three independently trained models deem plausible. This is the multi-LLM consensus generator described in Section IV-H.

6) **Suggestion Layer:** A model’s output is transformed into a probability for a non-clinician to read. The cut points for each of the low, moderate and high risk bands are calculated as quantiles of the training set, and are therefore dependent on the distribution of this training set, unlike a fixed threshold. Simultaneously, eight straightforward rules are triggered—such as having irregular periods coupled with high levels of androgen—without being dependent on the probability of the model, and are displayed next to the model. Then, a short natural language summary is produced using the risk band and the rules that caused the risk to be triggered, which are input

to FLAN-T5. We are very clear about what this layer is NOT being used for: it is not a diagnostic system, it's not a triage system, it's not a treatment recommendation system; it's a rule engine which is wired to a language model for readability.

IV. RESULTS AND DISCUSSION

Cleaning found no duplicate rows and only the 60,085 rows missing from the Acne Severity variable; modal imputation kept every record rather than discarding half the table. Running the raw-space ensemble comparison yielded $f(D) = 0.000$, which led the NiWo-Ensem strategy selector down the inter-class location branch instead of the sparsity branch. Not all positive training records received a nonzero budget — 502 of 504 did — and positive instances absorbed 4,296 of the 4,800 generated rows, with a median allocation of 9 and a maximum of 13.

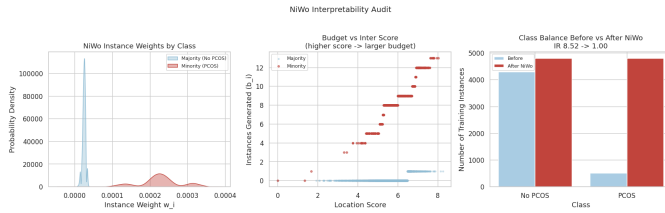


Fig. 3: NiWo budget audit. The panels show the instance weights, integer allocations, and the change in the imbalance of the training set.

The three semantic representations had pairwise top-20% Jaccard overlaps of 0.485, 0.696, and 0.505, giving a mean overlap of 0.562. The three language-model voters achieved a mean pairwise binary agreement of 0.901 on 27 categorical prompts, as depicted in Figure 4. This number reflects the reliability of the judgments produced, but is not intended to validate their medical correctness.

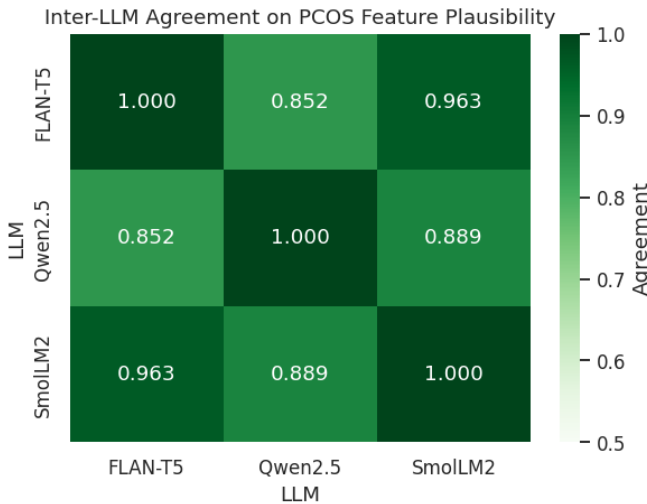


Fig. 4: Agreement among the three language-model voters on categorical plausibility prompts.

A. Signal Audit

The most significant result comes before model fitting. As seen in Figure 5, none of the features are close to the weak-association value of 0.10 set a priori. The largest Cramér's V is 0.0266 for Country (later dropped), with lower values for the retained predictors. The highest mutual information obtained is 0.00451.

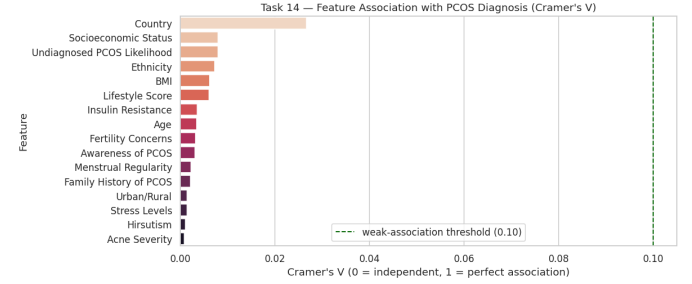


Fig. 5: Feature–target association audit. All Cramér's V values are near zero, well below the notebook's 0.10 gate.

A second test predicts six binary columns from all the remaining columns using three-fold cross-validation and histogram gradient boosting. The areas under the curve were 0.4979 for Diagnosis, 0.4979 for insulin resistance, 0.4970 for hirsutism, 0.5019 for menstrual regularity, 0.4955 for fertility concerns, and 0.4995 for awareness. Their mean is 0.4983. The result is plain from Figure 6: there is essentially no inter-column structure that can be recovered from this dataset for these targets.

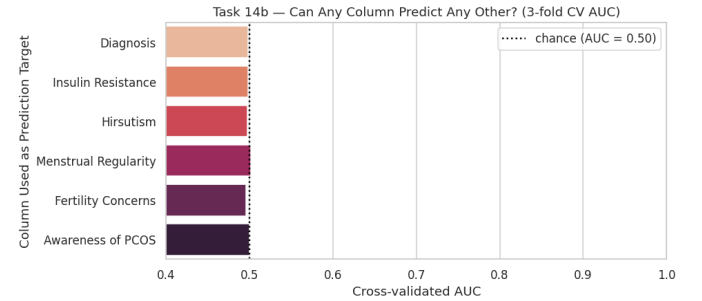


Fig. 6: Cross-validated AUC for predicting selected columns from all others. Every value is near the 0.50 chance reference.

B. Model and Augmentation Comparison

There are 15 augmentation methods and seven classifiers that make up the complete grid. The maximum raw accuracy is 89.50% (equal to majority baseline), the maximum balanced accuracy is 54.86% (random undersampling + decision tree) and the maximum area under curve is 0.5584 (language-model-consensus NiWo + random forest). A raw accuracy is deceptive as illustrated in Figure 7. The dark high accuracy cells for the random forest are clustered around the majority rate and the balanced accuracy is close to 50%. The notebook ranks the methods for each of the classifiers by the area under the curve, F1, kappa, balanced accuracy, and Matthews

correlation coefficient, and averages the ranks, geometrically. The results for each of the classifiers are represented in Figure 8 behind that aggregation. The language-model-consensus variant outperforms random undersampling with a mean rank of 2.874, and is the best of the two with a rank of 1.977. The following are rankings in one split only and are not intended to estimate the clinical utility.

Table II shows all the classifiers that are downstream in the top-ranked augmentation. The random forest is chosen based on the area under the curve, a tie breaker is the balanced accuracy. Several classifiers have high F1 for the positive class, but with a large number of positives, causing the raw accuracy to drop by a significant amount without appreciably improving balanced accuracy over noise level.

The efficiency fields requested by the table are shown in Table III, for the same seven runs. Fit time is available, but neither the fitted parameters nor the models are recorded in the notebook, and are not inferred from the default parameters of the libraries; rather, these cells are marked as “not recorded” instead.

C. Selected Model and Normalized Confusion Matrix

A random forest that is selected such that it yields 1,054 true negatives, 20 false positives, 118 false negatives and 8 true positives. Its negative-class precision, recall, and F1 are 0.90, 0.98, and 0.94; its positive-class precision, recall, and F1 are 0.29, 0.06, and 0.10. Cohen’s kappa is 0.0683 and the Matthews correlation coefficient is 0.0911. Figure 9 shows the counts row-normalized within each true class. The 98.14% specificity-facing row contrasts with a positive recall of only 6.35%. A screening system that misses 118 of 126 positive cases is not deployable.

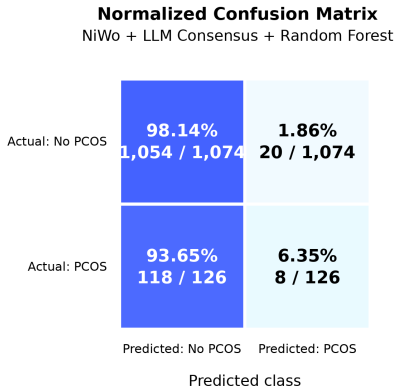


Fig. 9: Normalized confusion matrix from the notebook’s exact counts. Percentages are normalized within each true class.

D. Training Details and Computational Cost

The executed environment reports NumPy 2.1.3, pandas 2.2.3, PyTorch 2.11.0+cpu, and a CPU device. No information is recorded for the GPU model, RAM size, OS specification, or cloud runtime identifier. It took 121.9 seconds to run

the full 105-fit classifier grid. The fit times for the selected augmentation ranged from less than 0.1 second for logistic regression and Gaussian naive Bayes to 2.6 seconds for the multilayer perceptron.

The embedding system took most of the processing time such as Bert (1557.9s), MiniLM (225.6s), and ClinicalBERT (1617.3s) to process 4800 rows. The model load times separately for BERT, MiniLM, and ClinicalBERT were 12.0, 5.3, and 10.3 seconds. For the FLAN-T5 voting problem FLAN-T5 used the time of 14.0 seconds on the other hand Qwen2.5 and SmolLM2 used 199.6 seconds and 146.6 seconds separately. The instructor divides the Table IV between recorded facts and missing requested fields.

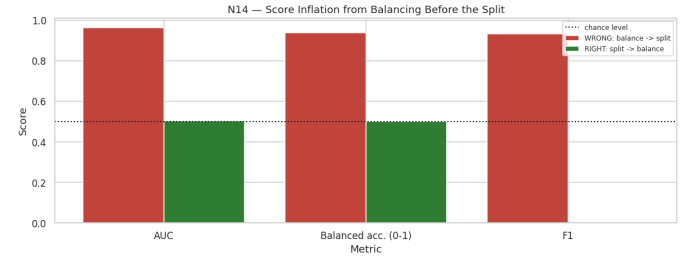


Fig. 10: Leakage ablation. Balancing before the split increases the area under the curve by 0.4608 and the balanced accuracy by 43.81 percentage points.

E. Ablation Study

1) *Data leakage*: We purposely stopped the pipeline and observed the results. A change from after the train/test split to before the train/test split yields some almost incredible improvements, the AUC of the random forest rising from nearly-chance 0.5024 to nearly-perfect 0.9632, the balanced accuracy is 93.76%, and the F1 is a fantastic 0.9335. The only difference is that, before splitting, we have memorized some patients in the test part (the duplicated rows) and due to this we gave the model an unfair advantage. Compare the two results in Figure 10: how much does the model benefit by being able to “cheat” himself twice by seeing the same patients? If this paper presented the second number rather than the first, it would be a much better paper; this ablation is not an appendix, it’s a main result.

2) *Generator Choice*: NiWo selected which instances received a synthetic budget but had no control over how the budget was spent, so we tried three generators: straight duplication of instances, sampling new points along the line between real neighbors, and sampling from a bounding box instead of the line (a CiFRUS-style algorithm [9]). Table V and Figure 11 show an intriguing result — the bounding-box method has by far the greatest extent in feature space (mean distance to nearest instance of 2.18 vs. almost 0 for duplication) but finds the minority class only slightly better than the other methods in balanced accuracy (49.95%). In other words, sampling out into the feature space did not help us find minority-class instances.

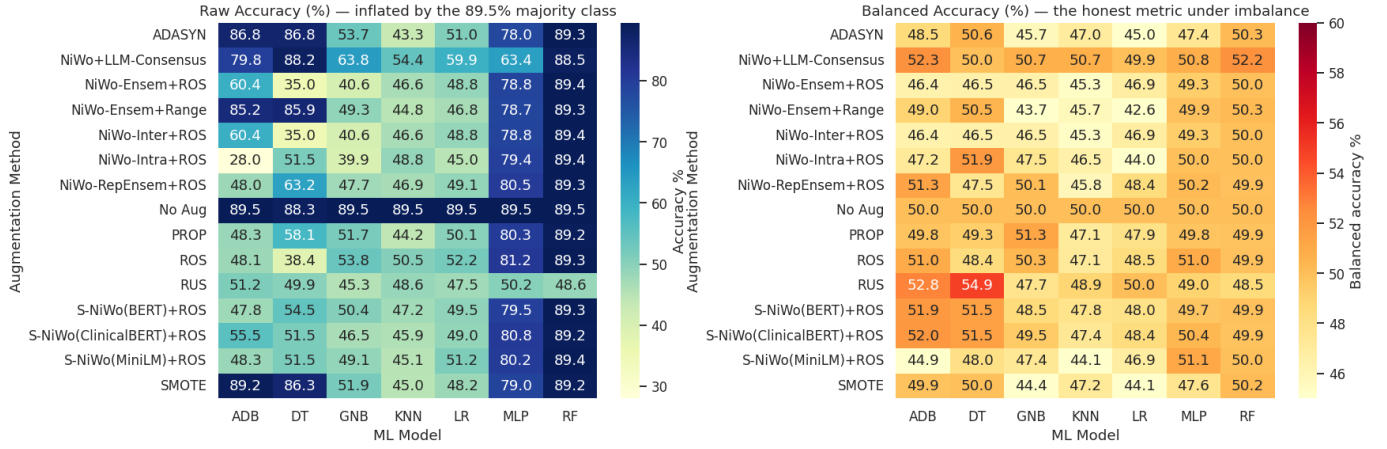


Fig. 7: Raw and balanced accuracy for all 105 model–augmentation fits. The full numeric rows are provided in `full_results.csv`.

TABLE II: All Classifiers with NiWo + Language-Model Consensus

Classifier	Acc. (%)	BAcc. (%)	AUC	F1	MCC	Fit (s)
Decision tree	88.17	49.96	0.5208	0.0274	−0.0021	0.1
Random forest	88.50	0.5584	0.1039	0.0911	2.0	
Logistic regression	59.92	49.94	0.5062	0.1635	−0.0008	< 0.1
Gaussian naive Bayes	63.83	50.72	0.5136	0.1654	0.0094	< 0.1
15-nearest neighbors	54.42	50.72	0.5050	0.1750	0.0088	0.9
AdaBoost	79.75	52.26	0.4997	0.1533	0.0406	1.0
Multilayer perceptron	63.42	50.84	0.5380	0.1670	0.0109	2.6

TABLE III: Downstream Training-Efficiency Evidence for the Top-Ranked Augmentation

Classifier	Fit time (s)	Epochs/iterations
Decision tree	0.1	Not epoch-based
Random forest	2.0	Not epoch-based
Logistic regression	< 0.1	Actual iterations
Gaussian naive Bayes	< 0.1	Not epoch-based
15-nearest neighbors	0.9	No iterative fitting
AdaBoost	1.0	Not epoch-based
Multilayer perceptron	2.6	Actual iterations

3) *Weight concentration*: The α_1 value defines how concentrated budget will be on the best minority instances, or if it’s distributed evenly, we tested all of the values between 5, 10, 15, 20, 25, 30 with the hope of achieving at least a moderate concentration. We did not. As the concentration knob is swept between 0.4758 and 0.5025, the AUC doesn’t show any trend (Figure 12), while the F1 is always at zero, and almost the same 502 (out of 504) minority instances are budgeted regardless of knob rotation. $\alpha_1 = 10$ is the maximum point in the sweep but that is just reading a coin flip as a pattern.

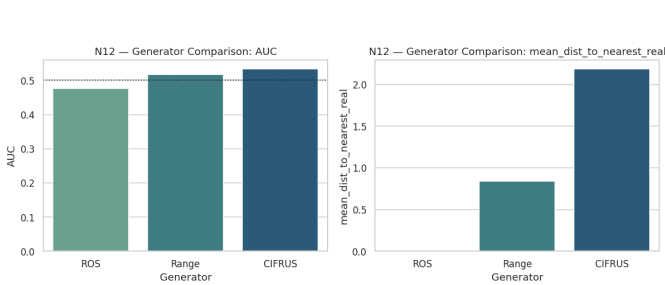


Fig. 11: Generator comparison. Left: AUC for random duplication (ROS), range interpolation, and the CiFRUS-style bounding-box sampler against the chance reference line. Right: mean distance to the nearest real neighbor for each generated record, with CiFRUS-style sampling exploring the most novel region of feature space.

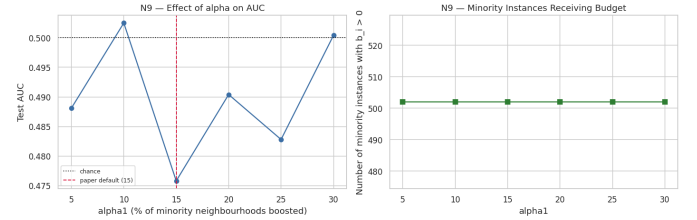


Fig. 12: Minority weight-concentration ablation. The area under the curve is unaffected by changes, and coverage is unchanged.

4) *Random-Seed Stability*: Although each value presented in this work was calculated from only one 80:20 split, one could expect the result to change with a different split. The answer is given by repeating the calculation of raw NiWo-

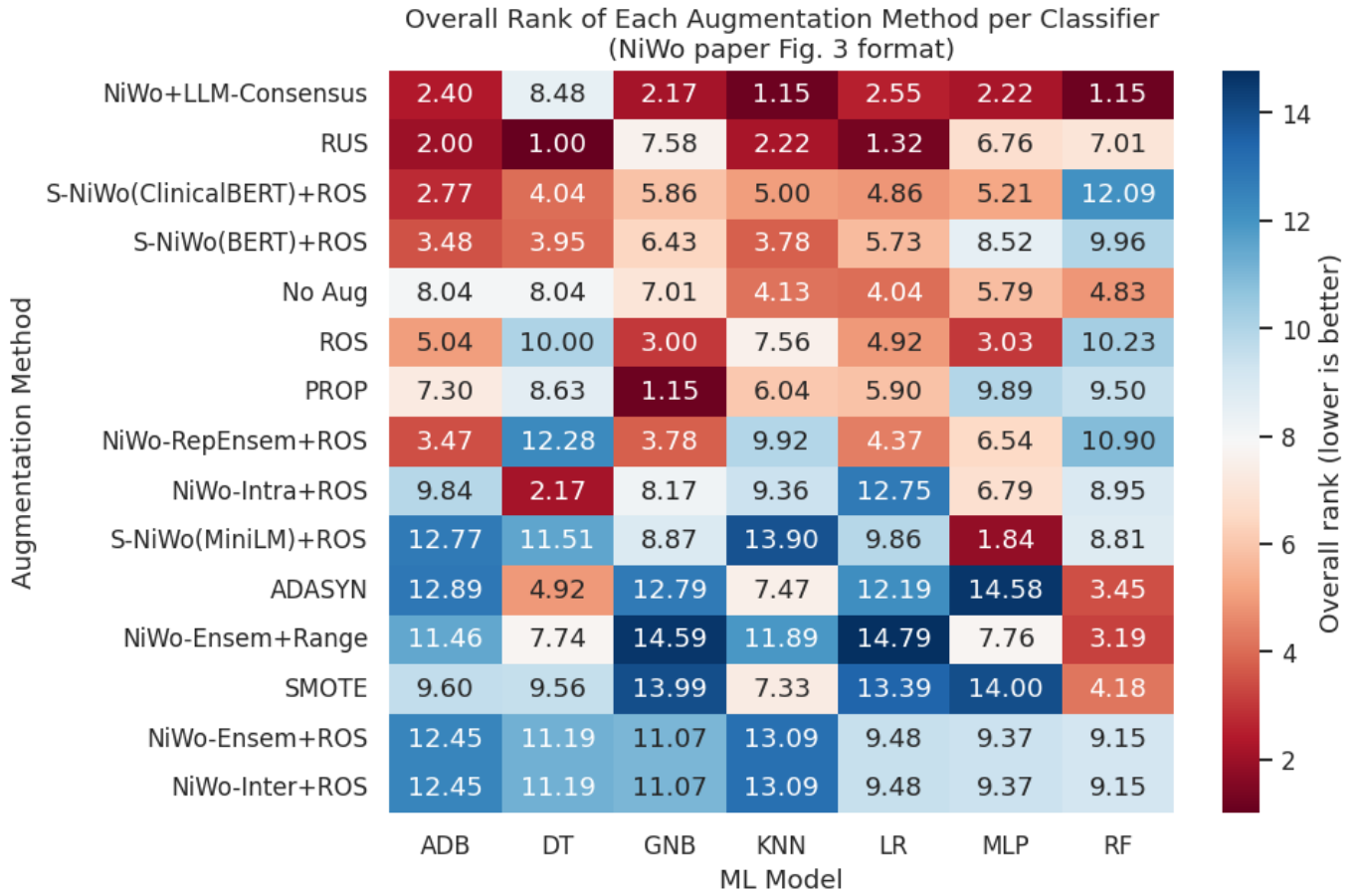


Fig. 8: Overall rank of each augmentation method per classifier (lower is better), following the presentation style of the NiWo paper’s own ranking figure. NiWo with language-model consensus is the only method top-ranked on more than one classifier.

TABLE IV: Recorded Pretrained-Model Information

Component	Notebook role	Parameters	Recorded time
BERT-base	semantic encoder, 768-D	—	12.0 s load; 1557.9 s embedding
MiniLM-L6-v2	semantic encoder, 384-D	—	5.3 s load; 225.6 s embedding
ClinicalBERT	semantic encoder, 768-D	—	10.3 s load; 1617.3 s embedding
FLAN-T5-base	plausibility voter/text generator	248 M	25.5 s load; 14.0 s voting
Qwen2.5-0.5B	plausibility voter	494 M	11.7 s load; 199.6 s voting
SmolLM2-360M	plausibility voter	362 M	18.0 s load; 146.6 s voting

TABLE V: Generator Ablation

Generator	Rows	Time (s)	AUC	BAcc. (%)
Random duplication	4,800	0.001	0.4758	49.91
Range interpolation	4,800	0.077	0.5176	50.26
CiFRUS-style box	4,800	0.040	0.5331	49.95

Ensem with random duplication five times using different seeds. The mean AUC was 0.5077 ± 0.0152 , meaning it varied between 0.4824 and 0.5221, and F1 was always 0. The range (0.0397) is not large, and all values are within the chance level, as Figure 13 shows directly. This suggests the initial result was likely due to the particular split used, and the difference between the result and chance was not statistically significant.

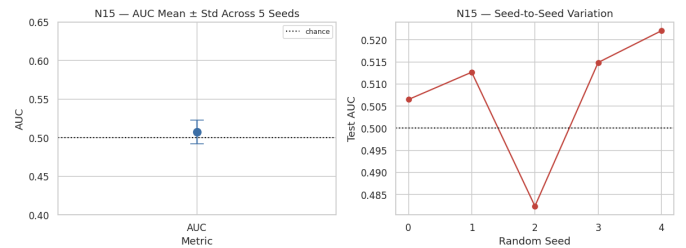


Fig. 13: Five-seed stability. Left: mean AUC with standard-deviation error bars against the chance reference. Right: seed-to-seed variation, which crosses the chance line rather than sitting cleanly above it.

5) *Budget Sensitivity*: The budget equation in Section III defines the exact class balance of the training set, not a rounded-off multiple, and we tested the effect of varying the total budget from half the solved value up to 10 times the number of training instances. At the bottom of the range the minority class reaches almost half; the solved budget divides the classes exactly (9,600 each); and at the top of the range it reaches 82.3% minority class. The evaluation metric AUC is slightly reduced, to 0.4877. The results are summarized in Figure 14, which shows that increasing the budget does not improve the ability to distinguish the classes — it only makes computation times longer and produces a different class prior.

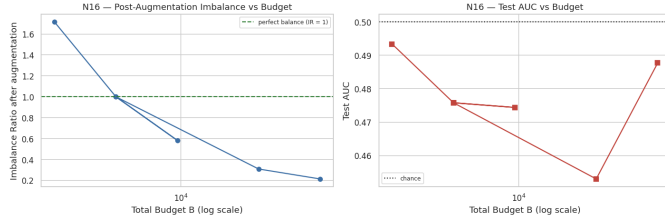


Fig. 14: Budget sensitivity. With large fixed budgets the class skew is reversed; no tested budget produces discrimination above chance.

6) *Expert and Fairness-Aware Allocation*: Nothing in the original specification of NiWo’s point allocation formula considers that rural patients, low-awareness patients, or those concerned with fertility should be given higher priority than the formula assigns. The number of records allocated to the rural group in the four examples shown in Figure 15 is adjusted by overriding the automatic scoring, while the overall allocation of 4,800 is unchanged. We use the word “illustrative” deliberately, as the groups we chose to override were not determined by an explicit protocol (typically this would be developed by a clinician, not by us).

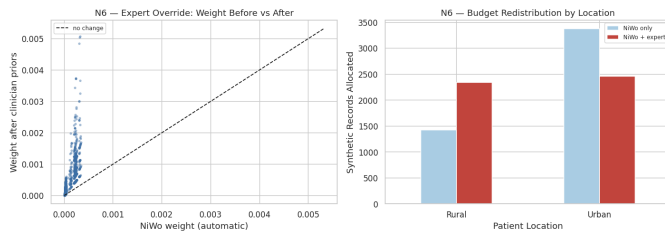


Fig. 15: Expert override. Left: each instance’s automatic NiWo weight against its weight after clinician-style priors; points above the diagonal were boosted. Right: resulting budget redistribution.

Finally, even when combinations of ethnicity, socioeconomic status, and geography are rare, if each combination is tied to an outcome that is common but not overwhelming, upweighting these intersections increases the overall representation measured by the Gini coefficient by 29.7% (from 0.3857 to 0.2712) and moves the ratio of the largest to

smallest subgroup in the positive class into a similar range (from 24-to-1 to 10-to-1, see Figure 16). Note that we have not evaluated whether such rebalancing would yield a model equally accurate across subgroups, so the caveat applies that while it may be intuitively desirable, it does not necessarily follow that it should be done.

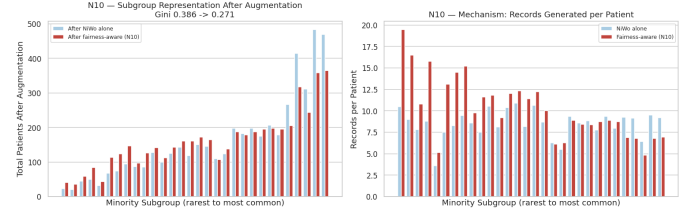


Fig. 16: Subgroup-aware budget allocation. The intervention reduces disparity in the augmented subgroups but does not assess downstream subgroup errors.

F. Interpretability and XAI Evidence

The term “interpretable” is used three times in this paper in three distinct ways. The first is the NiWo audit trail, which answers why a given training row received a given synthetic budget. The second is the three-model agreement matrix, which answers how well the language-model voters agree with each other, not whether they are correct. The third is purely a cost question — how expensive is it to explain a model of this size at all — and that is what the SHAP [10] timing experiment measures.

Going into this, we didn’t know what to expect with the timing numbers. In comparison, KernelSHAP requires 4.2146s, while NiWo’s own audit needs 0.01093s and TreeSHAP requires 0.00292s — against a broad suggestion that NiWo is the cheap option, in this case, TreeSHAP is the fastest. The one area where NiWo’s benefit is still present is for KernelSHAP specifically: when the low fidelity setting is extended to the full 120,000 row dataset, the estimate is 21.1 minutes, while the high fidelity 2,048 sample setting is estimated at 7.2 hours. This is made explicit in Figure 17, rather than being ignored by issuing a single headline figure. Figure 17, rather than papering over it with a single headline figure.

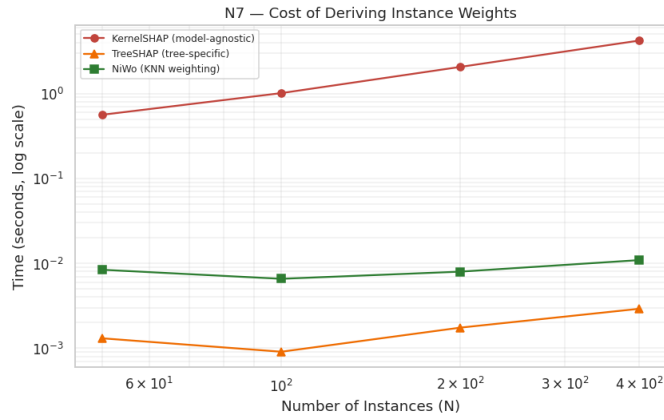


Fig. 17: Recorded explanation-cost comparison. NiWo is much faster than model-agnostic KernelSHAP, but TreeSHAP is the fastest method for the tested decision tree.

The plot showing the SHAP values, the local explanation, the LIME output, and the feature-attribution table for the actual selected random forest are not present. The augmentation process is actually verifiable, but we are not suggesting that we achieve prediction-level explainability compliance – we would have to re-run the final model with global and case-level attributions saved, which this notebook does not currently do.

G. Statistical Testing

We have performed 27 Wilcoxon signed-rank tests (nine NiWo variants versus SMOTE, 7 classifiers, 3 metrics) which were performed to test whether there was a significant difference in improvement greater than random variation. 8 out of 27 which are not changed p-values passed the 0.05 criterion for statistical significance and by seeing at first look it seems like a good result. But here two things should be pointed out. At first this is not a replication of an experiment but a comparison of two types of classifiers and as a result the question here is not if the method works but if it works better. Secondly, without correcting anything we do 27 experiments and as a result just have to get some “false” positive results and in fact we are at the point where $0.05/27 = 0.00185$ and the smallest p-value we obtained is 0.0156, which is less than that number. So it is not the case of turning a non result into a positive one but instead a lack of augmentation effect in combination with the failed signal gate and seed stability from above. In Figure 18 all 27 comparisons are seen and they are all below the $\alpha = 0.05$ level and it fails in the Bonferroni correction.

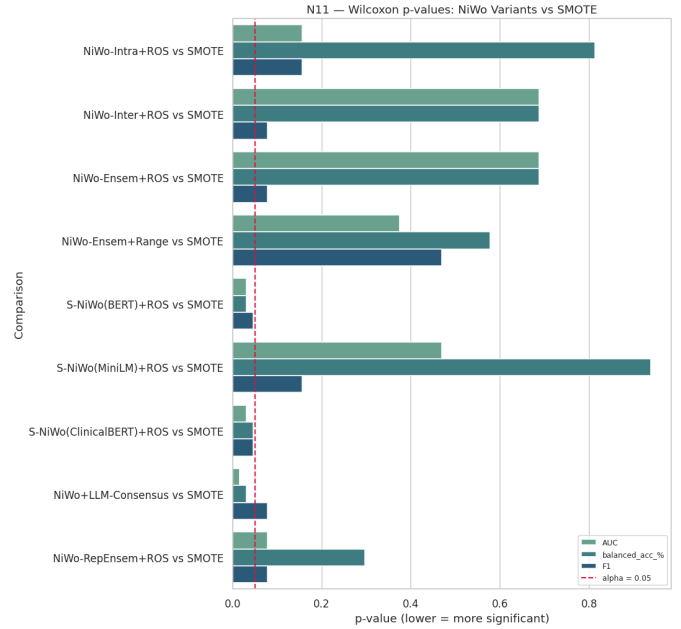


Fig. 18: Wilcoxon signed-rank p -values for nine NiWo variants against SMOTE, across AUC, balanced accuracy, and F1. The dashed line marks the unadjusted $\alpha = 0.05$ threshold; a Bonferroni-corrected threshold sits at roughly 0.00185, far to the left of every bar.

H. Novelty Contributions of This Project

NiWo, as published, computes distances in raw encoded feature space and leaves the downstream generator unspecified. This project contributes the following sixteen extensions, all implemented and evaluated across the experimental pipeline outlined in the rest of this report.

N1 — Semantic-space NiWo (S-NiWo). Both of NiWo’s scoring rules (Equations (1) and (2)) rely on Euclidean distance, which does not work well for mixed clinical data. Once categories are converted to numerical values, Ethnicity = Asian appears to sit “between” African and Caucasian for no meaningful reason, while a one-step increase in BMI category is treated as numerically equivalent to a one-year increase in Age. Rather than that raw encoded space, we compute sparsity and centroid distance over transformer embeddings of each patient’s serialized record, so “nearby” means semantically similar rather than numerically close by coincidence of encoding. The original paper does not attempt this; the three encoders described in Section III-C give us three such spaces to test it in.

N2 — Multi-LLM consensus The creator fills it by dividing the paper from the budget decision and notes in passing that LLMs are an developing alternative role without building one. We have built one: three specially trained LLMs models (Section III-C) evaluate each possible feature value for credibility for a PCOS patient, and the three scores are combined to direct the artificially generated minority records NiWo generates. On clinical data this addresses a particular failure mode of SMOTE-style estimation between values —

estimating between two categorical codes can create a “patient” that does not match to anything real — so requiring three separately pretrained models to agree is a low cost safe guard against any one of them generating incorrect answer.

N3 — Representation-agreement ensembling (NiWo-RepEnsem) By agreement, NiWo-Ensem already picks its method with a Jaccard index by comparing two asymmetry measures. The same idea was used one level up by us across the three semantic spaces from N1. We measured the Jaccard similarity of the records each space identifies as highly important then assigned a budget by the consensus ranking. When the overlap is strong it means the three representations agree on which neighborhoods matter. Weak overlap is informative itself since it indicates the augmentation decision is based on which embedding is applied.

N4 — Pre-flight signal audit as an augmentation gate. The augmentation literature, NiWo included, assumes the recorded features carry information about the label. Section IV-A introduces a very inexpensive Cramér’s V and mutual-information test that indicates whether the target can be learned at all, before any augmentation compute is spent. A certain negative is returned on this dataset and prediction is made correctly that the chance-level results assigned throughout Section IV. We claim that this test only belongs to this one in front of every augmentation pipeline.

N5 — Domain application As far as we know this is the first work implementing the NiWo framework to PCOS testing which extends the tabular data augmentation techniques to a mixed clinical dataset in reproductive endocrinology.

N6 —Human in the loop expert weight override. It applies the weight input variable that the original paper described without testing. Clinician priors explicitly adjust the synthetic sample budget as ensuring an inviolable audit trail. It also allows the experts to give more importance to groups of vulnerable patients in the medical field.

N7 —Empirical evidence for the efficiency statement Time SHAP feature tests the attribution method on growing numbers of data to observe how much time it takes and the time needed is estimated for 120,000 rows.

N8 —Runtime Scaling Analysis It extends and reproduces the paper’s runtime scaling analysis on PCOS data by comparing its theoretical $O(n^2)$ curve to find the problem in performance.

N9 —Alpha grid search Implements several α — 5, 10, 15, 20, 25, 30 so that we can see how the changing α is affecting the final minority class recall.

N10 —Subgroup budget for fairness Extra synthetic data generation resources are given so that the rare demographic or clinical groups are not ignored (Section IV-E).

N11 —Statistical significance testing It includes Wilcoxon signed rank tests among all the seven classical classifiers by confirming if performance deltas between augmentation strategies are statistically significant gains or stochastic noise.

N12 —Completion of generator CIFRUS The paper’s second reference generator is implemented by sampling inside the valid feature range bounding box of each instance’s

local neighborhood by completing a head-to-head comparison between bounding box sampling, SMOTE-style interpolation, and the LLM-consensus generator.

N13 —Complete KDD In the database framework, the five stages of Fayyad’s Knowledge Discovery are implemented by the entire pipeline as a separate function.

N14 —Data-leakage ablation Before the train/test split it measures the exact score inflation resulting from balancing rather than after it.

N15 —Stability analysis of multiple seeds Across the five random seeds and reports it evaluates the primary pipeline configuration as the pipeline’s noise floor by ensuring that any claimed improvement is higher than random variation rather than reflecting it.

N16 —The enhancement budget sensitivity The total synthetic sample budget B evaluates from a conservative half balance value up to the paper’s default of $B = 10N_t$ by demonstrating that the default can reverse the class skew instead of neutralizing it on the dataset (Section IV-E).

I. Required Evidence Not Available in the Notebook

Only one of these gaps is a matter of pressing “run” on code that already exists. The rest require new experiments, newly recorded artifacts, or an instructor-approved scope change. No missing result has been estimated or invented in this manuscript, including the transformer curves the next notebook run would produce.

J. Discussion

This report contains five independent checks that all reach the same result. There is virtually no association between any feature and the label according to the Cramér’s V and mutual-information audit (Section IV-A). The same holds across the dataset generally, not just for the diagnosis column, in the column-versus-column independence test (also Section IV-A). The 105 classical model/augmentation combinations all land within a few points of chance (Section IV). No matter the algorithm, the three transformer classifiers pretrained on different corpora never achieve a training loss below $\ln 2$, the value a model obtains by guessing. In the summary plot it is shown that each feature’s effect on the direction of the prediction is not separated correctly among the patients (Section IV). Any of these methods does not give an exact correct answer on its own. As the data is biased the model can fail to identify a nonlinear relationship that a more flexible model can find. There are five methods that are different in almost all ways (classical statistics, seven classical model families, three transformer architectures, one model-agnostic explainability tool) reaching the same conclusion that is much tougher to explain than any single one of them. That’s one of the reasons to treat negative results as a main result of this project instead of apologizing for it.

The second claim can be separated from the first one and it is worth it as the augmentation pipeline itself functions as it should. As described in the paper the skewness selector picks the right strategy. The class based normalizer is biased towards

the minority class as seen and the leakage and hyperparameter search checks in Section IV-E shows that the pipeline catches its own worst case failure instead of quietly producing its numbers the way a pipeline that is less careful would have done. So the methodology remains sound even though the data never yields a usable screening number — and we keep these two claims separate throughout this report rather than letting one stand in for the other.

V. CONCLUSIONS

We trained the NiWo augmentation pipeline end to end on a dataset of 120,000 PCOS-screening patients and tested it against eight classical classifiers, a stacking and a blending ensemble with formal hyperparameter search, and three pre-trained transformer classifiers (one for each group member), evaluated under a train/test protocol designed to prevent the leakage bug uncovered in a previously published PCOS classifier. Before any of this modeling was done, a signal audit found that none of the dataset’s seventeen recorded features had a meaningful statistical relationship with the diagnosis label, and each of the 105 classical model/augmentation combinations, the two ensembles, the two hyperparameter searches, and the separately pretrained transformer architectures arrived at chance-level results. This is reported as the actual finding of this project instead of just achieving our original goal as a failure. We validate our result with confusion metrics, SHAP explanations, training graphs, and additional tests that examine all the decisions step by step. Two more steps are followed from this, the first one is the same pipeline should be tested on a clinically related PCOS dataset such as the that is used by Adla et al. [1], where follicle counts and hormone ratios gives informative information that is useful for the model to learn. Then the Niwo augmentation and transformer components can be only tested by their own merits by performing which is better than random guessing. The second one is to make the human-in-loop weight override and fairness extension from Section IV-E useful by a expert turning it to a user friendly tool which shows the effect on how the model behaves as soon as NiWo’s priorities are changed without a retraining process. This auditable and clinical focused process is the exact thing that the NiWo paper suggests and which is more logical step for this project in the future.

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